

Problem

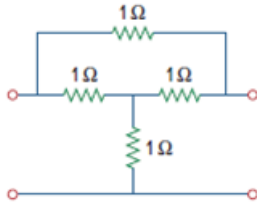
For the bridge circuit in Fig. 19.108, obtain:

(a) the z parameters

(b) the h parameters

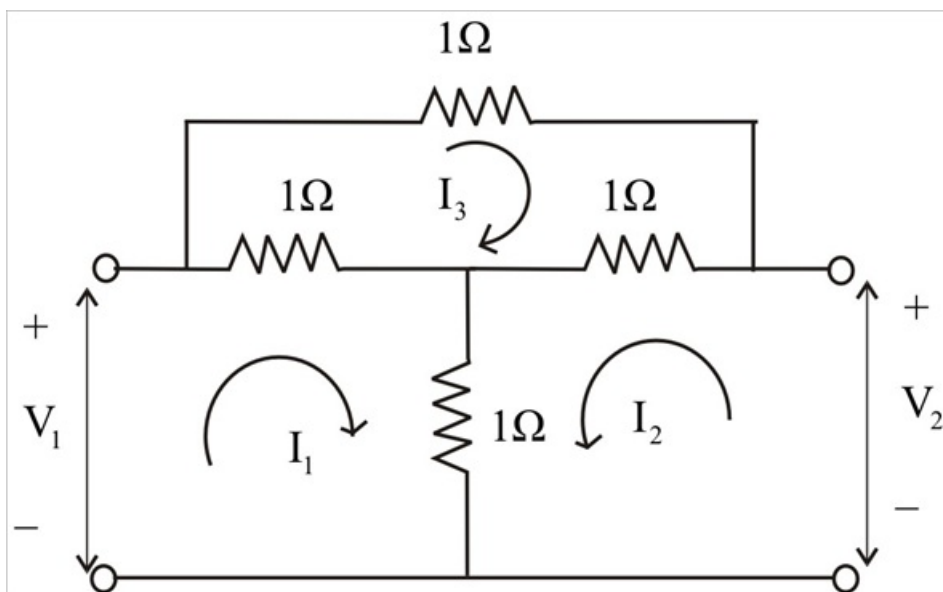
(c) the transmission parameters

Figure 19.108



Step-by-step solution

Step 1 of 6



Step 2 of 6

Using mesh analysis we have

$$\begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ 0 \end{bmatrix}$$

$$\Rightarrow 2I_1 + I_2 - I_3 = V_1 \quad -(1)$$

$$I_1 + 2I_2 + I_3 = V_2 \quad -(2)$$

$$-I_1 + I_2 + 3I_3 = 0 \Rightarrow I_3 = \frac{I_1 - I_2}{3}$$

Step 3 of 6

Using equations (1) and (2) we get

$$2I_1 + I_2 - \left(\frac{I_1 - I_2}{3}\right) = V_1$$

$$\Rightarrow 5I_1 + 4I_2 = 3V_1 \quad -(3)$$

$$\text{And } I_1 + 2I_2 + \left(\frac{I_1 - I_2}{3}\right) = V_2$$

$$\Rightarrow 4I_1 + 5I_2 = 3V_2 \quad -(4)$$

From equation (3) and (4) we have

$$\frac{1}{3} \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

Step 4 of 6

$$(a) [Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{5}{3} & \frac{4}{3} \\ \frac{4}{3} & \frac{5}{3} \end{bmatrix} \Omega$$

$$\left[\text{So, } \Delta Z = \frac{25}{9} - \frac{16}{9} = 1 \right]$$

Step 5 of 6

$$(b) [h] = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \frac{1}{Z_{22}} \begin{bmatrix} \Delta Z & Z_{12} \\ -Z_{21} & 1 \end{bmatrix} = \frac{1}{\left(\frac{5}{3}\right)} \begin{bmatrix} 1 & \frac{4}{3} \\ -\frac{4}{3} & 1 \end{bmatrix} = \begin{bmatrix} \frac{3}{5} \Omega & \frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} S \end{bmatrix}$$

Step 6 of 6

$$(c) [T] = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \frac{1}{Z_{21}} \begin{bmatrix} Z_{11} & \Delta Z \\ 1 & Z_{22} \end{bmatrix} = \frac{1}{\left(\frac{4}{3}\right)} \begin{bmatrix} \frac{5}{3} & 1 \\ 1 & \frac{5}{3} \end{bmatrix} = \begin{bmatrix} \frac{5}{4} & \frac{3}{4} \Omega \\ \frac{3}{4} S & \frac{5}{4} \end{bmatrix}$$

$X \rightarrow (2) \text{End}$